

THE SYMMETRIC RANK-ONE METHOD

Lecture 18: Presented by Jan van Vuuren



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The SR1 update formula (p.88)

The Symmetric Rank-1 update formula is due to the American mathematician and physicist William Cooper Davidon (1927–2013)

It corrects the first deficiency of Broyden's update rule (its inability to preserve symmetry of the Hessian matrix series of estimates $\mathbf{B}^0, \mathbf{B}^1, \mathbf{B}^2, \dots$) and is one of the most popular quasi-Newton update formulas among practitioners

Recall that a rank-1 correction term assumes the form $\Delta^k = \mathbf{u}^k (\mathbf{v}^k)^T$ and that the vector $\mathbf{u}^k$ is required to be specified as

$$\mathbf{u}^k = \frac{\mathbf{y}^k - \mathbf{B}^k \mathbf{s}^k}{(\mathbf{v}^k)^T \mathbf{s}^k}$$

in order to ensure that the update rule satisfies the Secant condition

$$\mathbf{B}^{k+1} \mathbf{s}^k = \mathbf{y}^k$$

Therefore, a rank-1 update formula necessarily takes the form

$$\mathbf{B}^{k+1} = \mathbf{B}^k + \frac{\mathbf{y}^k - \mathbf{B}^k \mathbf{s}^k}{(\mathbf{v}^k)^T \mathbf{s}^k} (\mathbf{v}^k)^T$$

for some n -vector $\mathbf{v}^k$.

The SR1 update formula (p. 88)

Theorem 3.7 (Characterisation of Hessian estimate symmetry preservation)

The quasi-Newton update rule

$$\mathbf{B}^{k+1} = \mathbf{B}^k + \frac{\mathbf{y}^k - \mathbf{B}^k \mathbf{s}^k}{(\mathbf{v}^k)^T \mathbf{s}^k} (\mathbf{v}^k)^T$$

preserves symmetry in the sequence $\mathbf{B}^0, \mathbf{B}^1, \mathbf{B}^2, \dots$ of Hessian matrix estimates if and only if the n -vector $\mathbf{v}^k$ is a scalar multiple of $\mathbf{y}^k - \mathbf{B}^k \mathbf{s}^k$

The simplest choice for the vector $\mathbf{v}^k$ is therefore $\mathbf{v}^k = \mathbf{y}^k - \mathbf{B}^k \mathbf{s}^k$

This yields the SR1 update rule

$$\mathbf{B}^{k+1} = \mathbf{B}^k + \frac{(\mathbf{y}^k - \mathbf{B}^k \mathbf{s}^k)(\mathbf{y}^k - \mathbf{B}^k \mathbf{s}^k)^T}{(\mathbf{y}^k - \mathbf{B}^k \mathbf{s}^k)^T \mathbf{s}^k}$$

or, by applying the Sherman-Morrison formula,

$$\mathbf{H}^{k+1} = \mathbf{H}^k + \frac{(\mathbf{s}^k - \mathbf{H}^k \mathbf{y}^k)(\mathbf{s}^k - \mathbf{H}^k \mathbf{y}^k)^T}{(\mathbf{s}^k - \mathbf{H}^k \mathbf{y}^k)^T \mathbf{y}^k}.$$

Symmetric Rank-1 method in pseudo-code (p. 89)

Algorithm 3.12 (The Symmetric Rank-1 method)

Input: An instance of a multivariate differentiable function $f : \mathbb{R}^n \mapsto \mathbb{R}$, an initial estimate $\mathbf{x}^0$ as to an approximate local optimum of f , an initial Hessian estimate $\mathbf{B}^0$ or an initial inverse Hessian estimate $\mathbf{H}^0$, as well as a search termination criterion

Output: An approximation of a local optimum of f close to $\mathbf{x}^0$ satisfying the termination criterion

Step 1 $k \leftarrow 0$

Step 2 If the search termination criterion is met, then output $\mathbf{x}^k$ and stop

Step 3 Compute the search direction $\mathbf{p}^k$ by Gauss-Jordan elimination to solve the system of linear equations $\mathbf{B}^k \mathbf{p}^k = -\nabla f(\mathbf{x}^k)$, or compute it directly as $\mathbf{p}^k = -\mathbf{H}^k \nabla f(\mathbf{x}^k)$

Step 4 Compute an appropriate step length α^k in the search direction $\mathbf{p}^k$ by performing an exact line search or a backtracking Armijo-Goldstein line search in respect of the function $f(\mathbf{x}^k + \alpha^k \mathbf{p}^k)$

Step 5 Compute the next search estimate as $\mathbf{x}^{k+1} \leftarrow \mathbf{x}^k + \alpha^k \mathbf{p}^k$

Step 6 $\mathbf{s}^k \leftarrow \mathbf{x}^{k+1} - \mathbf{x}^k$

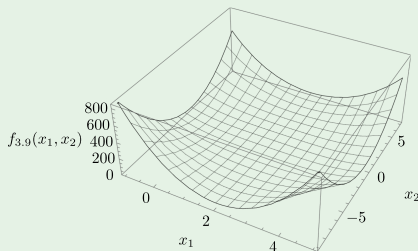
Step 7 $\mathbf{y}^k \leftarrow \nabla f(\mathbf{x}^{k+1}) - \nabla f(\mathbf{x}^k)$

Step 8 $\mathbf{B}^{k+1} \leftarrow \mathbf{B}^k + \frac{(\mathbf{y}^k - \mathbf{B}^k \mathbf{s}^k)(\mathbf{y}^k - \mathbf{B}^k \mathbf{s}^k)^T}{(\mathbf{y}^k - \mathbf{B}^k \mathbf{s}^k)^T \mathbf{s}^k} / \mathbf{H}^{k+1} \leftarrow \mathbf{H}^k + \frac{(\mathbf{s}^k - \mathbf{H}^k \mathbf{y}^k)(\mathbf{s}^k - \mathbf{H}^k \mathbf{y}^k)^T}{(\mathbf{s}^k - \mathbf{H}^k \mathbf{y}^k)^T \mathbf{y}^k}$

Step 9 $k \leftarrow k + 1$ and return to Step 2

Example 3.16 (pp. 89–92)

$$f_{3.9}(x_1, x_2) = 17 - 32x_1 + 24x_1^2 - 8x_1^3 + x_1^4 + 2x_2 + 5x_2^2 - 4x_1x_2^2 + x_1^2x_2^2$$



We apply Algorithm 3.12 (equipped with a Armijo-Goldstein line search with parameters $a = 0.4$ and $b = 0.9$) to minimise $f_{3.9}$, starting out from $\mathbf{x}^0 = [-1 \ 1]^T$

We are given an initial symmetric, positive definite Hessian estimate

$$\mathbf{B}^0 = \begin{bmatrix} 14 & -4 \\ -4 & 4 \end{bmatrix}$$

We work accurately to six decimal places and terminate the iterative process once the 2-norm $\|\nabla f_{3.9}(\mathbf{x}^k)\|_2$ drops below 0.05 for the first time.

Example 3.16 (pp. 89–92)

Our first step is to compute the gradient vector of $f_{3.9}$ as

$$\nabla f_{3.9}(x_1, x_2) = \begin{bmatrix} 2(x_1 - 2)(8 - 8x_1 + 2x_1^2 + x_2^2) \\ 2 + 2(5 - 4x_1 + x_1^2)x_2 \end{bmatrix}$$

During Iteration $k = 0$, the gradient vector is $\nabla f_{3.9}(\mathbf{x}^0) = [-114 \ 22]^T$

We compute the search direction by solving the system

$$\begin{bmatrix} 14 & -4 \\ -4 & 4 \end{bmatrix} \begin{bmatrix} p_1^0 \\ p_2^0 \end{bmatrix} = \begin{bmatrix} 114 \\ -22 \end{bmatrix}$$

The first search direction $\mathbf{p}^0 = [\frac{46}{5} \ \frac{37}{10}]^T = [9.2 \ 3.7]^T$ is obtained

A backtracking Armijo-Goldstein line search with parameters $a = 0.4$ and $b = 0.9$ yields the step length $\alpha^0 = 0.205891$

As a result, $\mathbf{s}^0 = \alpha^0 \mathbf{p}^0 \approx [1.894198 \ 0.761797]^T$

Therefore, $\mathbf{x}^1 = \mathbf{x}^0 + \alpha^0 \mathbf{p}^0 \approx [0.894198 \ 1.761797]^T$

Moreover, $\mathbf{y}^0 = \nabla f_{3.9}(\mathbf{x}^1) - \nabla f_{3.9}(\mathbf{x}^0) \approx [101.726656 \ -12.167764]^T$

Example 3.16 (pp. 89–92)

The output of the SR1 method coincides with that of Broyden's method during the first iteration ... the only difference comes in after the first Hessian update

The Hessian estimate is consequently updated to

$$\begin{aligned} B^1 &= B^0 + \frac{(\mathbf{y}^0 - B^0 \mathbf{s}^0)(\mathbf{y}^0 - B^0 \mathbf{s}^0)^T}{(\mathbf{y}^0 - B^0 \mathbf{s}^0)^T \mathbf{s}^0} \\ &= \begin{bmatrix} 14 & -4 \\ -4 & 4 \end{bmatrix} + \frac{1}{(78.255\,072)(1.894\,198) + (-7.638\,160)(0.761\,797)} \\ &\quad \times \begin{bmatrix} 78.255\,072 \\ -7.638\,160 \end{bmatrix} [78.255\,072 \quad -7.638\,160] \\ &\approx \begin{bmatrix} 57.001\,0114 & -8.197\,154 \\ -8.197\,154 & 4.409\,667 \end{bmatrix} \end{aligned}$$

Since

$$\|\nabla f_{3.9}(\mathbf{x}^0)\|_2 \approx 116.103\,402 > 0.05,$$

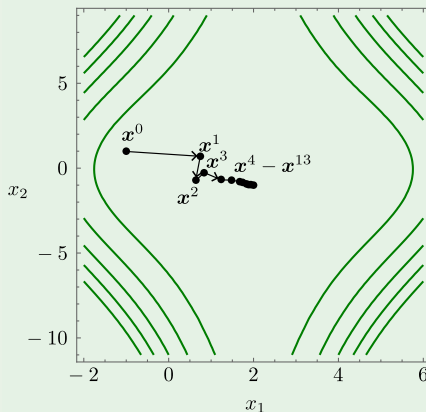
Algorithm 3.12 does not terminate immediately after the first iteration ...

Example 3.16 (pp. 89–92)

k	$\mathbf{x}^k$	$\nabla f(\mathbf{x}^k)$	$\mathbf{B}^k$	$\mathbf{p}^k$	α^k	$\mathbf{s}^k$	$\mathbf{y}^k$	$\ \nabla f(\mathbf{x}^k)\ _2$
0	$\begin{bmatrix} -1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} -114 \\ 22 \end{bmatrix}$	$\begin{bmatrix} 14 & -4 \\ -4 & 4 \end{bmatrix}$	$\begin{bmatrix} 9.2 \\ 3.7 \end{bmatrix}$	0.206	$\begin{bmatrix} 1.894 \\ 0.762 \end{bmatrix}$	$\begin{bmatrix} 101.727 \\ -12.168 \end{bmatrix}$	116.103
1	$\begin{bmatrix} 0.894 \\ 1.762 \end{bmatrix}$	$\begin{bmatrix} -12.273 \\ 9.832 \end{bmatrix}$	$\begin{bmatrix} 57.001 & -8.197 \\ -8.197 & 4.410 \end{bmatrix}$	$\begin{bmatrix} -0.144 \\ -2.497 \end{bmatrix}$	1	$\begin{bmatrix} -0.144 \\ -2.497 \end{bmatrix}$	$\begin{bmatrix} 3.118 \\ -11.598 \end{bmatrix}$	15.726
2	$\begin{bmatrix} 0.750 \\ -0.735 \end{bmatrix}$	$\begin{bmatrix} -9.155 \\ -1.766 \end{bmatrix}$	$\begin{bmatrix} 71.639 & -5.373 \\ -5.373 & 4.954 \end{bmatrix}$	$\begin{bmatrix} 0.168 \\ 0.539 \end{bmatrix}$	1	$\begin{bmatrix} 0.168 \\ 0.539 \end{bmatrix}$	$\begin{bmatrix} 4.014 \\ 2.915 \end{bmatrix}$	9.324
3	$\begin{bmatrix} 0.919 \\ -0.196 \end{bmatrix}$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
11	$\begin{bmatrix} 1.959 \\ -1.022 \end{bmatrix}$	$\begin{bmatrix} -0.086 \\ -0.047 \end{bmatrix}$	$\begin{bmatrix} 2.122 & 0.207 \\ 0.207 & 2.046 \end{bmatrix}$	$\begin{bmatrix} 0.039 \\ 0.019 \end{bmatrix}$	1	$\begin{bmatrix} 0.039 \\ 0.019 \end{bmatrix}$	$\begin{bmatrix} 0.081 \\ 0.041 \end{bmatrix}$	0.098
12	$\begin{bmatrix} 1.998 \\ -1.003 \end{bmatrix}$	$\begin{bmatrix} -0.005 \\ -0.005 \end{bmatrix}$	$\begin{bmatrix} 2.042 & 0.116 \\ 0.116 & 1.943 \end{bmatrix}$	$\begin{bmatrix} 0.003 \\ 0.003 \end{bmatrix}$	1	$\begin{bmatrix} 0.003 \\ 0.003 \end{bmatrix}$	$\begin{bmatrix} 0.004 \\ 0.005 \end{bmatrix}$	0.007
13	$\begin{bmatrix} 2.000 \\ -1.000 \end{bmatrix}$	—	—	—	—	—	—	—

Example 3.16 (pp. 89–92)

Broyden's method returns $\mathbf{x}^{13} = [1.999\,812 \quad -1.000\,046]^T$ as approximate minimum of $f_{3.9}$; Compare this to the true minimum is $\mathbf{x}^* = [2 \quad -1]^T$



Although the SR1 method preserves symmetry of the Hessian estimates by design, the update rule suffers from the disadvantage that it does not preserve positive definiteness of these estimates

It is left as an exercise to show that during iteration $k = 3$ of Example 3.16, for instance, the positive definiteness of the sequence $\mathbf{B}^0, \mathbf{B}^1, \mathbf{B}^2, \dots$ of Hessian matrix estimates is momentarily lost, but then the SR1 method manages to regain this property again during the next iteration

It can, however, be shown that no rank-one update rule can preserve both symmetry and positive definiteness of the Hessian estimates

For that, a rank-two update rule is required.